

Chapter 6

Hemodynamic Attenuation of Attention Heads

6.1 Introduction

Earlier chapters used blood flow as data, training models to read hemodynamic signals. This chapter uses blood flow as a design principle: the way the brain decides which vessels to keep is used to decide which attention heads a transformer keeps.

In Chapter 4, a trained transformer carried its causal computation in two of forty heads, but this was found only after training, by ablating each head in turn, and the number of heads to keep had to be chosen by hand. This chapter asks whether a model can find the number of heads it needs during training, without being given it. We introduce a rule for this, hemodynamic attenuation, and test it on a task whose minimal number of heads is known.

6.2 Background

6.2.1 Blood supply

Four properties of cerebral blood supply motivate the rule:

1. Supply follows need: local blood flow rises with neural activity, signalled in part by astrocytes [2]. Each head is given a valve $g_h \in [0, 1]$.
2. A vessel is dispensable when collateral vessels can supply its tissue [3]. A head is valued by what is lost when it closes and the other heads compensate, its collateral value.
3. Maintaining a vessel has a metabolic cost [4]. Each open head has a fixed price; there is no target number of heads.
4. Unneeded vessels regress gradually [5, 6], and collaterals can enlarge when a route is blocked [3]. Closing heads fade out over 100 steps and can be reopened.

The biology motivates the design; no claim is made about the brain.

6.2.2 Astrocytes and attention

Kozachkov et al. [1] showed that neurons and one astrocyte can approximately compute self-attention. With hidden activity $h = \varphi(Wx) \in \mathbb{R}^m$ and output synapses $H \in \mathbb{R}^{d \times m}$, each scaled by an astrocyte process (their Eq. 6),

$$\ell = (H \odot \tilde{P}) h + x, \quad \tilde{P}_{i\alpha} = 1/p_{i\alpha}. \quad (6.1)$$

Calcium equalises across the astrocyte, so every process takes the same value (their Eq. 8) and $H \odot \tilde{P} = H/p$ with $p = \frac{1}{m} \sum_j \varphi(k_j)^\top \varphi(q_t)$. With Hebbian weights $H = \frac{1}{m} V \varphi(K)^\top$, the output for token t is (their Eq. 11)

$$\ell_t = \frac{V \varphi(K)^\top \varphi(q_t)}{\sum_j \varphi(k_j)^\top \varphi(q_t)} + x_t \approx \sum_i \frac{e^{k_i^\top q_t}}{\sum_j e^{k_j^\top q_t}} v_i + x_t, \quad (6.2)$$

since $\varphi(a)^\top \varphi(b) \approx e^{a^\top b}$. The neurons compute the numerator of the softmax and the astrocyte its denominator.

6.2.3 Head pruning

Michel et al. [10] remove heads by a gradient score, and Voita et al. [11] learn head gates with an L_0 penalty. Optimal Brain Damage [7] scores a parameter by the loss increase when it is removed; Optimal Brain Surgeon [8] by the increase after the remaining parameters are re-fitted. ZipLM [9] applies the latter to whole heads of trained language models. All of these prune after training, to a size chosen by the user.

6.3 Methods

6.3.1 Task

The memory has $N = 16$ slots, each holding a random item c_j . A query specifies a position p , and the target is the items at R fixed offsets (Figure 6.1a):

$$\text{target}(p) = (c_{p+\delta_1}, \dots, c_{p+\delta_R}), \quad \text{slots modulo } N. \quad (6.3)$$

Items are redrawn for every example, so the model must learn where to attend.

A head returns, for each query, one weighted average of the items: one linear equation in the R unknown items. Recovering R items needs R independent equations, so the task needs $k^* = R$ heads, and with $k < R$ heads the lowest achievable loss is

$$\text{loss}(k) = \left(1 - \frac{k}{R}\right) L_{\text{triv}}, \quad (6.4)$$

where L_{triv} is the loss of predicting the mean. Duplicate heads give the same equation and count once. Measured losses agree with Equation (6.4) to within 1% (Figure 6.1c).

The task counts as solved when the loss is below $0.02 L_{\text{triv}}$. Trained models sit hundreds of times below this threshold, and losing one needed head raises the loss by about L_{triv}/R , so results are unchanged for any threshold from 0.01 to 0.1.

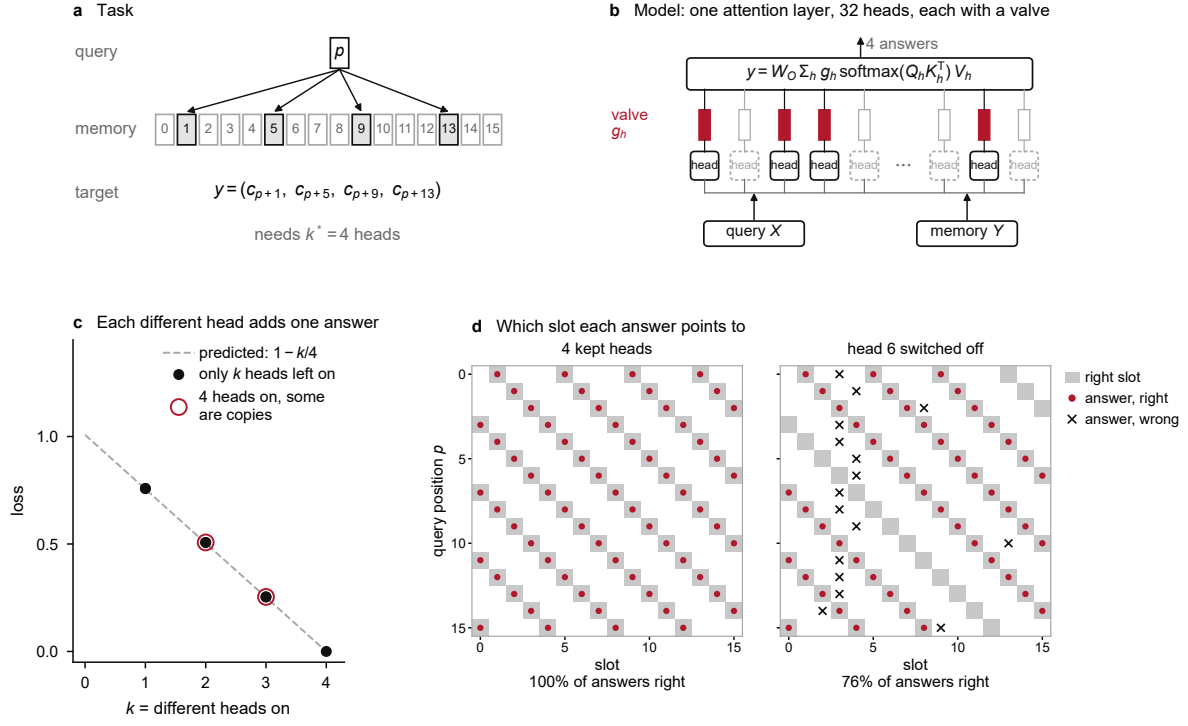


Figure 6.1: Task and model. **a**, A query carries a position p ; the target is the items stored 1, 5, 9, 13 slots ahead, so 4 heads are needed. **b**, One attention layer with 32 heads, each with a valve g_h ; filled valves are open. **c**, Loss with k different heads on a trained 4-head model, against the prediction $1 - k/4$. **d**, The slot each answer points to: all correct with the 4 kept heads, 76% with head 6 switched off.

6.3.2 Model

Matrices have one column per token, following [1]. The model is one cross-attention layer with $H = 32$ heads and no MLP, reading queries X and memory Y . Head h computes

$$E_h = \exp\left(\frac{1}{\sqrt{d_k}} (W_K^h Y)^\top W_Q^h X\right), \quad D_h = \text{diag}(\mathbf{1}^\top E_h), \quad O_h = W_V^h Y E_h D_h^{-1}, \quad (6.5)$$

where $E_h D_h^{-1}$ is the softmax. Each head is scaled by its valve before the heads are combined:

$$\hat{T} = \sum_{h=1}^H g_h W_O^h O_h = (W_O \odot P) O, \quad P_{i,(h,c)} = g_h. \quad (6.6)$$

A head with $g_h = 0$ contributes nothing and receives no gradient, so a model with k open valves is exactly a k -head model.

The right-hand form of Equation (6.6) is the astrocyte gain of Equation (6.1), constant within each head's block of W_O rather than over a whole domain. The layer therefore contains two gains of the same form (Table 6.1):

$$\hat{T} = \sum_h g_h W_O^h W_V^h Y E_h D_h^{-1}, \quad (6.7)$$

where D_h^{-1} is the per-query gain the astrocyte computes in [1] and g_h is the per-head gain set by our rule. The correspondence is algebraic only.

	Astrocyte gain [1]	Head valve
Matrix form	$H \odot \tilde{P}$, one value $1/p$	$W_O \odot P$, one value g_h per head
Set by	pooled input activity	collateral value relative to a price
Time scale	every token	over training
Function	softmax normalisation	selection of heads

Table 6.1: Two gains on blocks of synapses.

6.3.3 Hemodynamic attenuation

All valves start open. After a warm-up of 10% of training, every 25 steps the rule (1) computes the collateral value of every open head, (2) closes the lowest-valued head if its value is below the price $0.03 L_{\text{triv}}$, and (3) reopens the highest-valued closed head if its value exceeds twice the price. A closing valve decreases by 1/100 per step.

The collateral value is the increase in the best-fit loss when head h is removed and the other open heads re-fit their output weights:

$$v_h = \min_W L(\text{without } h) - \min_W L(\text{all open heads}). \quad (6.8)$$

With Z the stacked outputs of the open heads on a batch, $A = \frac{1}{n} Z Z^\top$, $W^* = \frac{1}{n} T Z^\top A^{-1}$, $M = A^{-1}$ and J the rows of head h ,

$$v_h = \frac{1}{d_{\text{out}}} \text{tr}(W_J^* (M_{JJ})^{-1} W_J^{*\top}), \quad (6.9)$$

one inverse for all heads. This is the block Optimal Brain Surgeon score [8, 9]; the contribution is its use during training, with a price, a gradual fade, and reopening.

The re-fit is what makes a copy worthless. If $z_2 = z_1$ and the target is z_1 , removing head 2 without re-fitting raises the loss by $0.25 \text{var}(z_1)$ when the weights are $(0.5, 0.5)$; with re-fitting head 1 takes weight 1 and $v_2 = 0$. Computing the values added about 5% to training compute. In one layer A is the exact curvature; deeper models would need a layer-wise approximation [9].

6.3.4 Baselines and evaluation

All methods share the model, task, training and runs. Published methods, run inside our training: ZipLM-style [9] (the same score applied once, then fine-tuning), Michel et al. [10], and Voita et al. [11]. Controls remove one part of the rule: no fade, no re-fit (the Optimal Brain Damage score [7]), and no score (random choice).

We report exact count (open heads over the last 40% of training equal k^*) and never broke (after first falling below the solved threshold, the loss never exceeds it). Experiments use $k^* = 2, 4, 8$ with 10 seeds each, plus runs with duplicated heads, runs with random head failure, and a two-layer induction task [12].

6.4 Results

Count and stability. Hemodynamic attenuation kept exactly k^* heads in 29 of 30 runs and never broke the model in 30 of 30; no other method achieved both (Table 6.2). Without the fade the model broke in every run; without the re-fit it kept too many heads; with random choice it broke in 24 of 30 runs.

Method (30 runs each)	Exact count	Never broke
Hemodynamic attenuation	29/30	30/30
ZipLM-style [9]	24/30	0/30
Michel et al. [10]	28/30	0/30
Voita et al. [11]	9/30	0/30
No fade	30/30	0/30
No re-fit	2/30	30/30
No score (random)	30/30	6/30

Table 6.2: Exact count and stability on the retrieval task ($k^* = 2, 4, 8, 10$ seeds each).

Duplicated heads. With every head starting as a copy of another, the rule kept 4 heads and never both members of a pair; without the re-fit, 8 to 15 heads and 4 to 7 pairs were kept (Figure 6.2d).

Backup heads. If each head fails with probability p , a head’s value with B heads open is

$$v(B) = \frac{1-p}{R} \Pr [\text{Binomial}(B-1, 1-p) \leq R-1], \quad (6.10)$$

and the rule should keep the largest B with $v(B)$ above the price. This prediction, made before the experiments, matched 21 of 24 runs exactly and the rest to within one head (Figure 6.2e).

Induction. In a two-layer attention-only model with $8 + 8$ heads, trained to predict the token that followed an earlier copy of the current token, the minimal circuit is one previous-token head and one induction head [12]. Hemodynamic attenuation kept $2 + 1$ heads in 10 of 10 runs, one more than the minimum, and never broke the model; the kept heads had the expected roles in all 10 runs, and switching off the induction head reduced next-token accuracy on the repeat to zero (Figure 6.3). On this task every method kept the same $2 + 1$ heads, so only stability separated them: the method of Michel et al. broke the model in all 10 runs, while the controls without re-fit or with random choice did not.

Cost. Final loss was 2 to 7 times the dense model’s, well below the threshold, using 0.31 to 0.45 of its head-steps. The saving is nominal, since closed heads are still computed.

6.5 Discussion

A rule built from how vessels are maintained and pruned recovered a known minimal circuit during a single training run, without being given its size and without the model leaving the solved regime. Removing any one of its components lost either the count or stability.

The tasks are small and the main model has one layer, which is what makes the answer known; the results do not yet extend to large language models. The collateral value is exact only for one layer, and the compute saving requires skipping closed heads in code. The biology motivates the rule but is not evidence for it.

Two experiments would connect this chapter to the rest of the thesis and have not yet been run: applying the rule to qSA-Cross, to test whether it keeps the two causal heads found in

Chapter 4, and to the Cross-Site Transformer on WildPPG, to test whether the same heads are kept across all sixteen leave-one-subject-out folds.

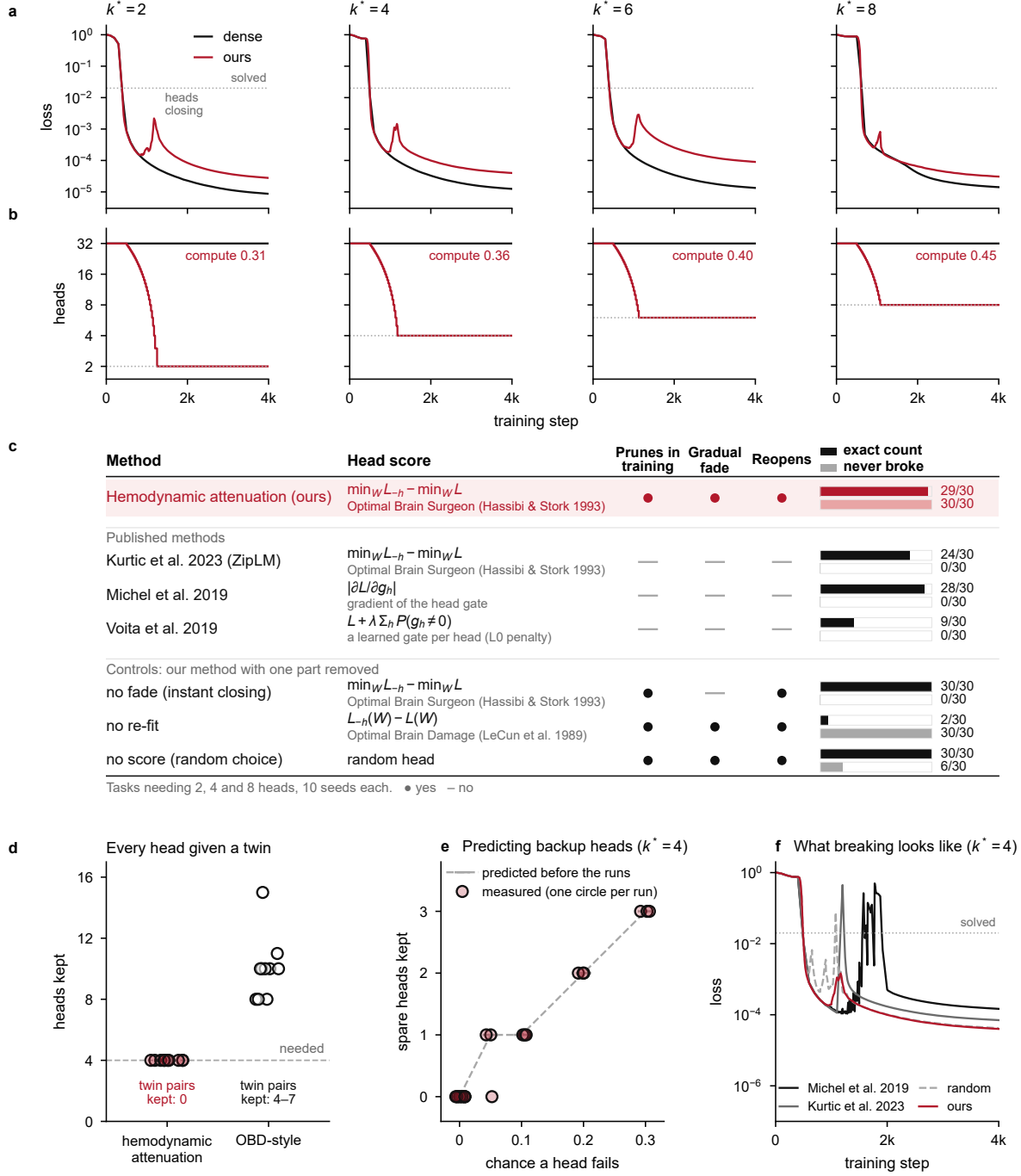


Figure 6.2: Hemodynamic attenuation during training. **a**, Loss of the dense model (black) and with hemodynamic attenuation (red). **b**, Open heads; dotted line, k^* . **c**, All methods on the same 30 runs. **d**, Runs in which every head starts with an identical copy. **e**, Spare heads kept under random failure, against the prediction. **f**, One run per method; only hemodynamic attenuation stays below the solved threshold.

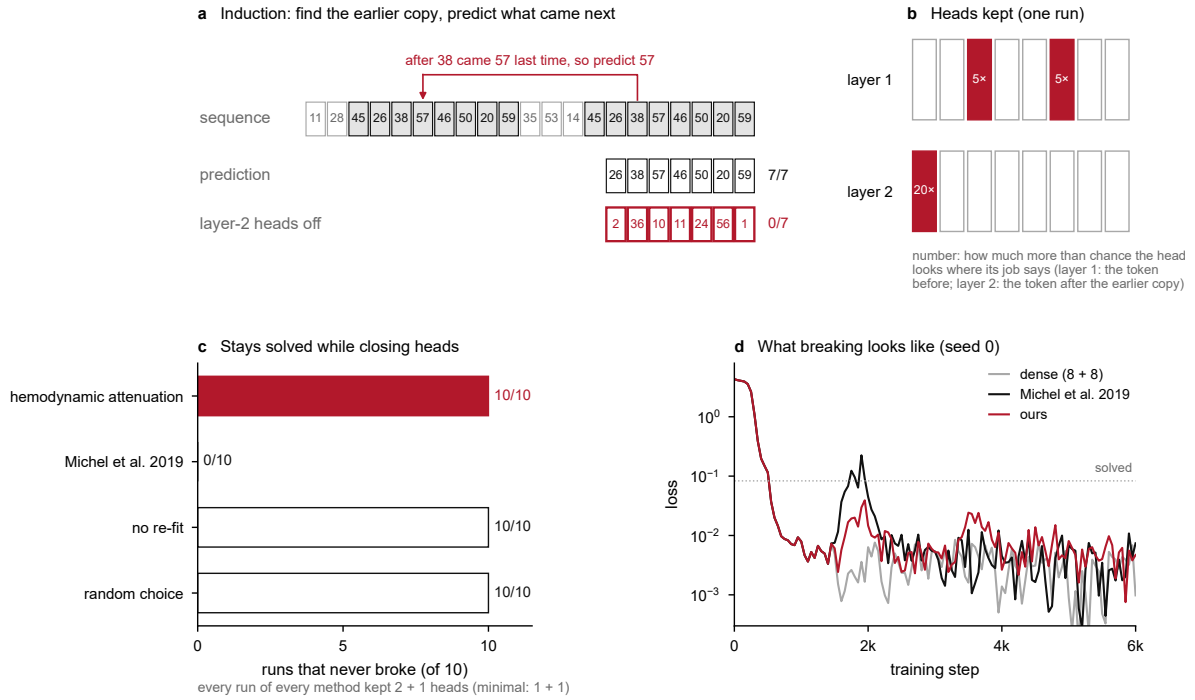


Figure 6.3: Two-layer induction. **a**, An example sequence and the model’s predictions on the repeat, with all kept heads (7/7 correct) and with the layer-2 head switched off (0/7). **b**, Heads kept in one run; numbers give how much more than chance each head attends to the position its role requires. **c**, Runs that never broke, 10 per method, all with the same settings. Every run kept 2 + 1 heads. **d**, Loss during training for one seed.

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